Wentao Ni

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Education

- **University of California San Diego** — San Diego, CA
  PhD, Computer Science and Engineering — *Sep. 2024 – Present*
- **Shanghai Jiao Tong University** — Shanghai, China
  B.S. in Computer Science and Technology, ACM Class, Zhiyuan College; GPA: 86.1/100.0 — *Sep. 2020 – June. 2024*

Publications

- **JUNO: Optimizing High-Dimensional Approximate Nearest Neighbour Search with Sparsity-Aware Algorithm and Ray-Tracing Core Mapping**

  Zihan Liu, **Wentao Ni**, Jingwen Leng, Yu Feng, Cong Guo, Quan Chen, Chao Li, Minyi Guo, Yuhao Zhu

  ASPLOS 2024.

- **VQ-LLM: High-performance Code Generation for Vector Quantization Augmented LLM Inference**

  Zihan Liu, Xinhao Luo, Junxian Guo, **Wentao Ni**, Yangjie Zhou, Yue Guan, Cong Guo, Weihao Cui, Yu Feng, Minyi Guo, Yuhao Zhu, Minjia Zhang, Jingwen LEng, Chen Jin

  HPCA 2025.

Research Experience

- **A Heterogeneous Top-K Token Selection Framework for Long Context LLM** — UCSD
  Advised by Prof. Jishen Zhao, Prof. Zhijian Liu — *Feb 2025 - Current*
  - Developed a predictive cluster selection method leveraging current-layer hidden states to estimate next-layer queries for IVF-based retrieval.
  - Implemented CPU+GPU hybrid processing for top-k token selection on top of FAISS, maximizing hardware utilization.
  - Built a dynamic workload scheduler using Integer Linear Programming (ILP) to minimize end-to-end latency under given system constraints.

- **High-performance Code Generation for Vector Quantization** — Shanghai Jiao Tong University
  Undergraduate Researcher, advised by Prof. Jingwen Leng — *Feb 2024 - June 2024*
  - We design and implement an efficient fused VQ kernel generation framework. We introduce a software abstraction called codebook cache to optimize codebook access efficiency and support the integration of VQ with various computations.
  - Additionally, we provide adaptive heuristics to tailor parameter selection in our optimizations to diverse VQ configurations. Our optimizations achieve an average latency reduction of 46.13% compared to unoptimized versions.

- **Optimize High-Dimensional ANNS with Ray-Tracing Core** — Shanghai Jiao Tong University
  Undergraduate Researcher, advised by Prof. Jingwen Leng — *June 2022 - May 2023*
  - We design a threshold-based selective algorithm to rapidly filter out the unnecessary search points leveraging the sparsity and spatial locality and I propose a mapping for our algorithm to run on the RT core.
  - We study how to generalize the existing kNN-RT core mapping to ANN search with arbitrary dimensions, in aspects of approximation method, metrics and system design, and propose JUNO, an end-to-end high-dimensional ANN search engine with both algorithmic enhancement and optimized hardware mapping.

- **Combining Graph & Tensor Transformation with Scheduling via Compilers** Duke University
  Research Assistant, advised by Prof. Yiran Chen *Sep. 2022 - Dec. 2023*
  - Due to the emergence of new forms of computation like State Space Models, researchers need to manually design CUDA kernels for these computations.
  - I explore TVM and EinNet to combine graph-level and tensor-level transformation on tensor expressions, along with the scheduling space, and implement it based on TVM te expression.

- **Search Efficient Network Architectures for LLM with Linear Complexity** Duke University
  Research Assistant, advised by Prof. Yiran Chen *Sep. 2022 - Dec. 2023*
  - Linear complexity architectures like State Space Models (SSM) are efficient in memory and computation. I used NAS-based distillation to search linear complexity architecture to substitute attention mechanism in transformers. I proposed a method to measure the complexity of generated architectures.

- **Improve Quantization by Searching Configurations** Microsoft Research Asia
  Intern, advised by Zhenhua Han, Yuqing Yang *Sep. 2023 - Dec. 2023*
  - Latest quantization research mainly focuses on improving quantization methods, neglecting the different sensitivity of weights in groups or layers.
  - I construct a search space of different granularity, dynamically assigning number of bits to each part of the weights based on profiled sensitivity information, to gain a better performance among the memory and accuracy trade-off.
  - With the help of sparse matrix kernels of previous works, hybrid quantization can be perfectly supported.

Projects

- **Mx* Compiler**
  - A Compiler from Mx* language (C++ like) to RV32I Assembly, with static optimizations on LLVM IR, implemented in Java.
  - Front-end parser and lexer use Antlr library. Middle-end LLVM IR generation, back-end RV32I assembly codegen, static optimizations are all handwritten.
  - Static optimizations include Static Single-Assignment Form (SSA), Sparse Conditional Const Propagation (SCCP), Aggressive Dead Code Elimination (ADCE) and loop inline.
- **RISC-V CPU**
  - A Tomasulo RISC-V CPU with iCache and branch predictor with 2-bit saturating counter, implemented in Verilog RTL.
- **Pytorch-like library**
  - Follow the course 10-414/714: Deep Learning System Course, CMU.
  - Implemented two levels. Python-level: forward and backward computation of operators, nn.modules, weight initializations, optimizers, dataloaders etc. Numpy-level: NDArray (handling memory, shape, strides and offset) with both CPU and CUDA backend.

Awards and Honors

- **Mathematical Contest In Modeling** 2021
  ***Outstanding Winner**, one of 36 outstanding winners out of 26112 teams*
- **The 45th ICPC Yinchuan Regional Programming Contest** 2021
  ***Gold Medal**, Rank: 19/491* *Yinchuan, China*
- **CCPC 2020 Weihai Regional Programming Contest** 2020
  ***Gold Medal**, Rank: 18/384* *Weihai, China*
- **The 45th ICPC Nanjing Regional Programming Contest** 2020
  ***Gold Medal**, Rank: 9/548* *Nanjing, China*

- **SJTU Zhiyuan Honors Program Scholarship** 2020, 2021
- **SJTU ACM Class Scholarship** 2020
- **National Olympiad in Informatics** 2019
  ***Bronze Medal**, Rank: 188/263* *Guangzhou, China*
- **National Olympiad in Informatics Winter Camp** 2019
  ***Silver Medal**, Rank: 152/381* *Guangzhou, China*

Part-time Working Experience

- **Teaching Assistant** 2022 - 2023
  Shanghai Jiao Tong University *Shanghai, China*
  - Data Structure (CS1951) Mar. – June 2022
  - Principle and Practice of Computer Algorithms (CS1952) June – Aug. 2022
  - Advanced Compiler (CS2965) Mar. – June 2023
    * Design a coding assignment about loop transformation on Polyhedral, based on SCoP format.

Programming Skills & Interests

- **Languages**: Python, C++, Java, Verilog, C, Objective-C, Matlab, LaTeX
- **Frameworks and Libraries**: Pytorch, TVM, CUDA, Pandas, Numpy, Matplotlib, Tensorflow, Flask
- **Mathematics** : Mathematical Analysis, Probability, Measure Theory, Advanced Algorithm
- **Interests** : Machine Learning System, Compiler, Computer Architecture